

ANSH VERMA

CSE (AI/ML) STUDENT | Software Engineering | AI / Generative AI | Backend & Full-Stack Development

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PROFESSIONAL SUMMARY

B.Tech Computer Science and Engineering (AI/ML) student and early-career software engineer building across frontend, backend, and AI: TypeScript, React, and Next.js interfaces, Fastify and PostgreSQL REST APIs, and Gemini-powered Generative AI applications. Hands-on experience in testing, verification, debugging, and Docker through hackathon projects, with Python and Git/GitHub fundamentals.

EDUCATION

B.Tech, Computer Science and Engineering (Artificial Intelligence & Machine Learning)

2025–2029

ABES Engineering College, Ghaziabad, Uttar Pradesh | CGPA: 7.16

TECHNICAL SKILLS

Programming: Python, TypeScript, JavaScript (ES6+), C++, SQL

Backend & Systems: Node.js, Fastify, RESTful APIs, PostgreSQL Transactions, Rate Limiting, Idempotency, Authentication

Databases: PostgreSQL (Exclusion Constraints, GiST Indexing), MongoDB, SQL Schema Design

Frontend Development: React, Next.js (App Router), Vite, Tailwind CSS, Semantic HTML5, CSS3, Responsive Design, Accessibility

AI & Developer Tooling: Machine Learning, Generative AI, Google Gemini API, Prompt Engineering, Web Speech API, Mutation Testing, Git, GitHub, Docker, Docker Compose

Core Computer Science: Data Structures & Algorithms, System Invariants, Relational Concurrency Control, Integration Testing, Verification, Debugging

PROJECTS

DevPartner AI – AI-Powered Developer Workflow (IBM Bob 2.0 Hackathon)

React, TypeScript, Vite, Node.js, Git/GitHub, AI-Assisted Development

- Contributed to the testing and verification layer of a team-built workflow that makes code changes safer through impact analysis, human approval, isolated patches, and an accept/rollback decision.
- Implemented invariant-checking and verification logic in TypeScript (harness.ts) to validate proposed code changes against identified invariants.
- Built React components (InvariantPanel, VerificationPanel, CounterexamplePanel) that display invariants, verification results, and adversarial counterexamples.
- Supported mutation-testing and adversarial-testing workflows and debugged verification behavior so failing cases were visible to reviewers.

HealthBuddy AI – AI Pre-Consultation Assistant (Hackathon Round 2 Prototype)

Next.js 16, React 19, TypeScript, Tailwind CSS, Gemini API, Web Speech API, REST APIs

- Developed a human-in-the-loop assistant that turns natural patient conversations into structured case sheets and doctor-ready summaries, without diagnosing or prescribing.
- Integrated the Google Gemini API through Next.js REST API routes for adaptive questioning in English, Hindi, and Hinglish, with an Offline Adaptive Clinical Engine for demo mode.
- Implemented live clinical information extraction, completeness checking, and triage/red-flag detection, with voice and text input via the browser Web Speech API.
- Built a doctor review console with editing, notes, digital verification, and audit-ready transcripts across the onboarding-to-summary flow.

TableKeeper – Restaurant Reservation Backend API

TypeScript, Fastify, PostgreSQL, Node.js, REST API, Docker, Docker Compose

- Engineered a Fastify and PostgreSQL JSON REST API in which a booking succeeds only when its database transaction commits.
- Enforced reservation consistency with PostgreSQL exclusion constraints, locking, and transactional commits to block conflicting confirmed two-hour reservations.
- Implemented restaurant-scoped idempotent requests, signed email-token customer verification (SMTP), confirmation-code authorization, and rate limiting.
- Validated with PostgreSQL 16 integration tests, migration upgrade tests, and TypeScript type checking; packaged with Docker Compose.

CERTIFICATIONS

- Gen AI NASSCOM** — SFI | 2026
- Getting Started with Artificial Intelligence** — IBM SkillsBuild | 2026
- Lab: Troubleshoot Your Code Using IBM Bob** — IBM SkillsBuild | 2026
- Adobe University Hackathon 2026, Round 1 Online Assessment (MCQ, Coding)** — Adobe | 2026
- HTML (ID: 96148998) and CSS (ID: 9595138)** — Simplilearn | 2025
- AI-ML Project Learning Workshop** — APPWARS Technologies (ABESEC) | 2026

ACHIEVEMENTS

- IBM Bob 2.0 Hackathon:** participated and contributed testing and verification to DevPartner AI.
- Internship Common Aptitude Test (iCAT), 2026:** scored 63, All India Rank #835, eligible for internship.
- Participated in Adobe University Hackathon 2026 (Round 1) and DECODE SIH 2026 (OSCode); built the HealthBuddy AI Round 2 prototype.